

ROMA TRE · A.Y. 2026/27

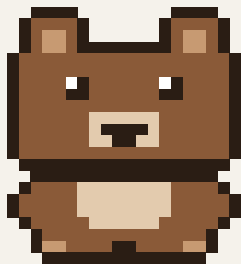
Policy Evaluation Lab

Lecture 0: Course presentation and syllabus

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PART 1

The course



About the course

- **Policy Evaluation Lab** – Master's degree (LM), first semester
- Scientific area: ECON-02/A (Applied economics)
- 60 hours, 9 CFU, taught in English
- Timetable: Monday, Thursday, and Friday at 5.00pm (Room 20)
- Course website (slides, datasets, exercises, announcements):
 - <https://pel.carusomatteo.it>
 - register with your university e-mail
- Official announcements are also posted on Moodle (available soon)



Why the need for a *policy*?

- In classical economics, State intervention is not needed: markets work by themselves
- This is rarely true
- We often need policies to:
 - ensure fairness and avoid inequality
 - reallocate and redistribute resources and endowments

Markets alone cannot ensure this

- We also need policies to:
 - improve markets themselves (e.g. antitrust policies)
 - fix negative externalities (e.g. pollution)

Policies to fix **market failures**



Why the *evaluation*?

- We can apply a policy, but then we want to understand:
 - whether it was the right policy
 - whether it could have been done better
 - how to improve it in the future
- Did it have the effect we were hoping for?
 - Did the health reform actually improve hospitals? Did people get better?
 - Did the minimum wage decrease employment? Or did it increase labour supply instead?
- We will never see what would have happened **without** the policy (the **counterfactual**), but we can make inference about it

This is what **causal inference** is about



Why a *lab*?

- We want to use **real-world data** to see how to evaluate policies
- We will make use of software for econometric analysis
 - the choice will depend on your background: from Excel to R
- We will see together how researchers and academics use this software to evaluate policies, provide new evidence on social dynamics, and better understand society as a whole
- Each week, one of the lectures will focus on a **class discussion** of a specific policy:
 - we will look at the data, discuss it, and think together about how it could have been done better

Seminars and discussions in class



Learning outcomes

By the end of the course you will be able to:

1. **Frame** a **policy question** in terms of **potential outcomes** and **counter-factuals**
2. **Recognise selection bias** and explain why **correlation is not causation**
3. **Choose** the right **research design** for a given policy and data availability
4. **Estimate treatment effects** using several econometric methods (e.g. RCTs, matching, DiD, IV, RDD and synthetic control)
5. **Assess** the **identifying assumptions** and run the standard **validity checks**
6. **Read, replicate** and **critically discuss** an applied evaluation study



PART 2

Programme



Programme (I)

Week	Topic	Lab / case study
1	Causality and potential outcomes	Selection bias; software set-up
2	Regression for policy evaluation	OLS, controls, robust and clustered SE
3	Randomised experiments	Tennessee STAR; balance checks, ITT
4	Selection on observables	Matching, propensity score, IPW
5	Difference-in-differences I	Card and Krueger (1994), minimum wage



Programme (II)

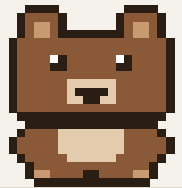
Week	Topic	Lab / case study
6	Difference-in-differences II	Event studies, staggered adoption
7	Instrumental variables	Angrist and Krueger (1991); LATE
8	Regression discontinuity	Sharp and fuzzy RDD, bandwidth choice
9	Synthetic control	Abadie et al. (2010), California Prop. 99
10	Wrap-up	Report presentations and exam review

The order may change slightly during the semester: check the website.



PART 3

Exam



Assessment

- **Report** on a policy intervention of your choice
 - either an empirical analysis in R with real-world data, or an argument based on explanation and evidence
 - the policy can be one you assess as **successful** or **unsuccessful**
 - either way, you have to argue your thesis properly, using **solid evidence**
- **Presentation** of your report in the last week of the course (week 10)
- **Oral exam**
 - written instead, if the number of students does not allow us to complete the orals on time

Argue your thesis with **solid evidence**



PART 4

Materials



Reading list

Main textbooks

- Angrist J.D., Pischke J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Cunningham S. (2021). *Causal Inference: The Mixtape*. Yale University Press.
Free online: <https://mixtape.scunning.com>

Further reading

- Angrist J.D., Krueger A.B. (1999). "Empirical strategies in labor economics". *Handbook of Labor Economics*, Vol. 3A, ch. 23.
- OECD (2025). *Implementation Toolkit for the OECD Recommendation on Public Policy Evaluation*. OECD Publishing, Paris. [Link \(PDF\)](#)



Software

- Labs are run in **R** (open source, so you can work on it yourselves) and **Excel**
 - R + RStudio: free, <https://posit.co/download/rstudio-desktop>
- **Stata** will be used only for a few examples
- Bring a laptop to the lab sessions
- Interactive simulations of each method:
<https://carusomatteo.it/econvis.html>



The course website: registration

1. Go to <https://pel.carusomatteo.it>
2. Register with your **university e-mail**, student ID (*matricola*), degree programme and a password of your choice
3. Enter the **course access code** given in class
4. From then on, log in with your e-mail and password

ACCESS CODE

Written on the board in class



The course website: a tour

- **Home:** announcements, timetable, location and office hours
- **Materials:** everything, week by week
 - slides are posted **before** each lecture; datasets and code **after**
 - some material (e.g. solutions) is released at a later date
- **Exercises:** problem sets for each method and interactive practice on EconVis
- **Exam:** assessment rules and exam sessions
- **Resources:** textbooks, software guides and datasets
- **Account:** your details and password

Check the **announcements** regularly



PART 5

Let's start

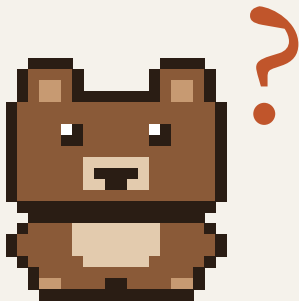


Correlation is not causation

- Two variables can move together without one causing the other
- Let's look at some examples together:
 - **Spurious correlations:** <https://www.tylervigen.com/spurious-correlations>

Correlation is not causation





Questions?

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